

Characterization of Galaxies via Optical and Infrared Surveys

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Abstract

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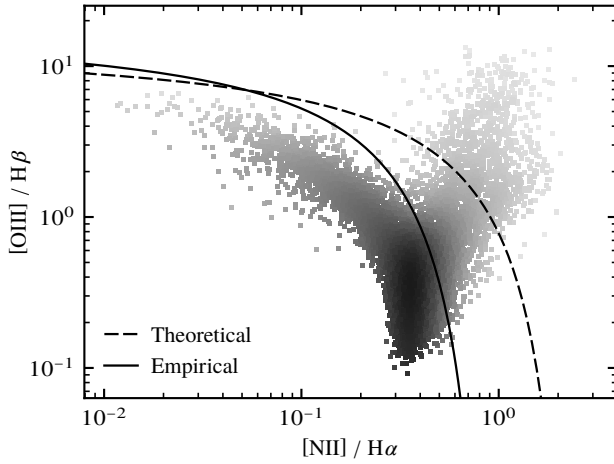


Figure 1

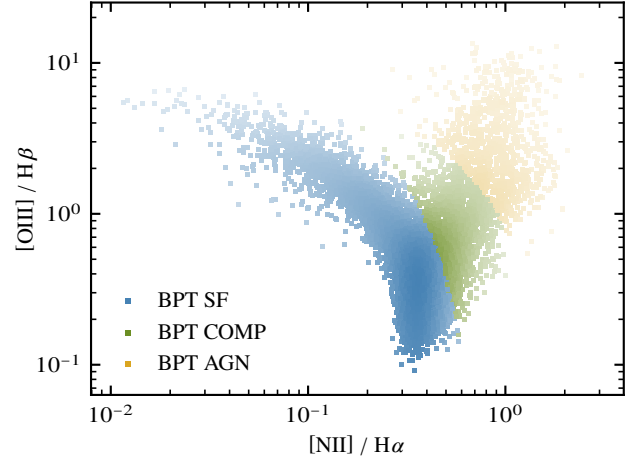


Figure 2

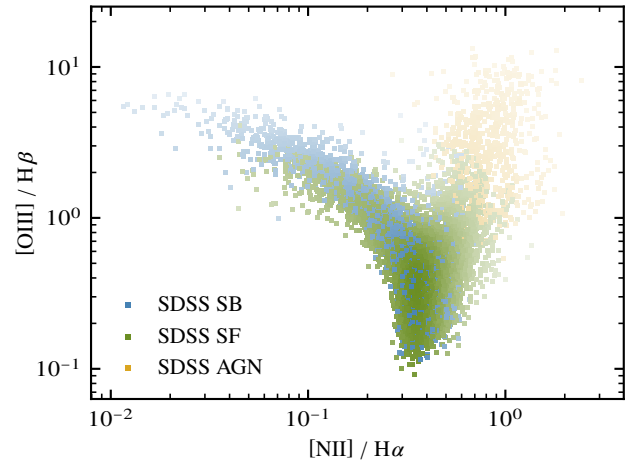


Figure 3

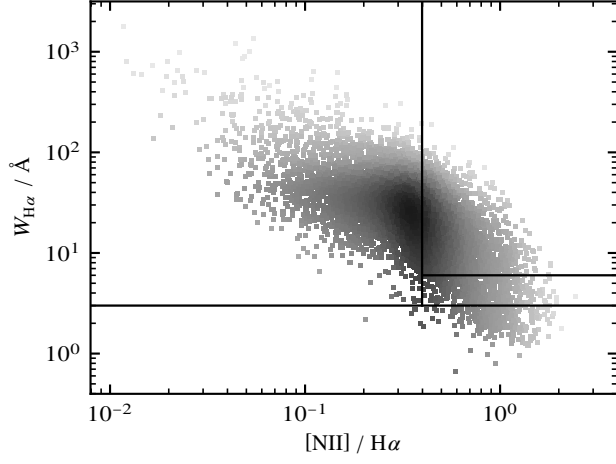


Figure 4

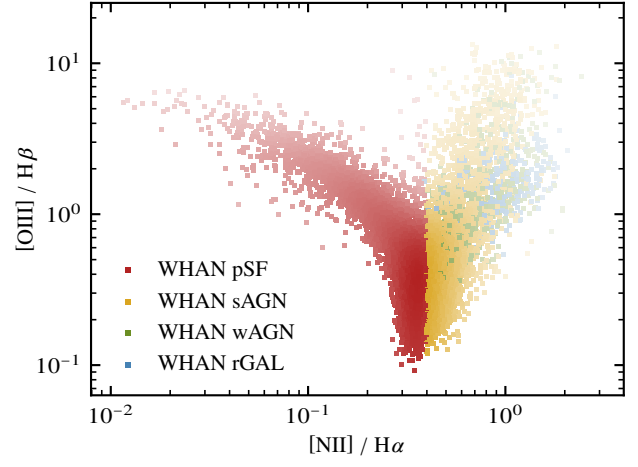


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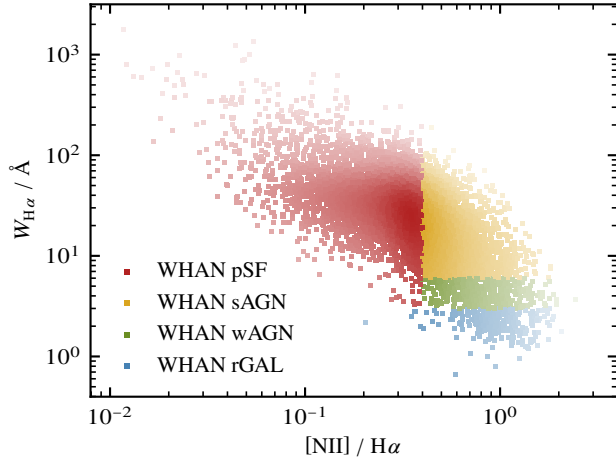


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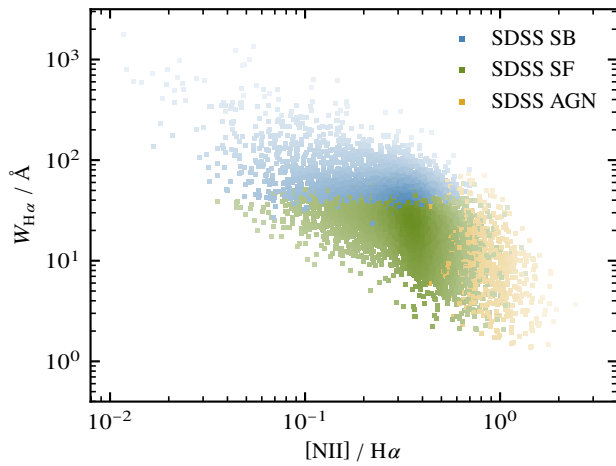


Figure 6

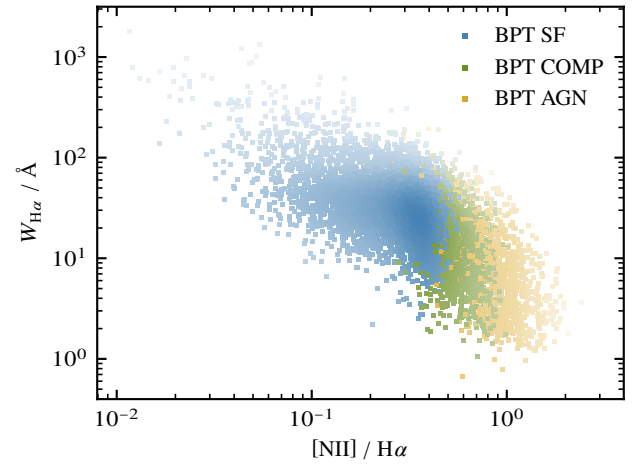


Figure 8

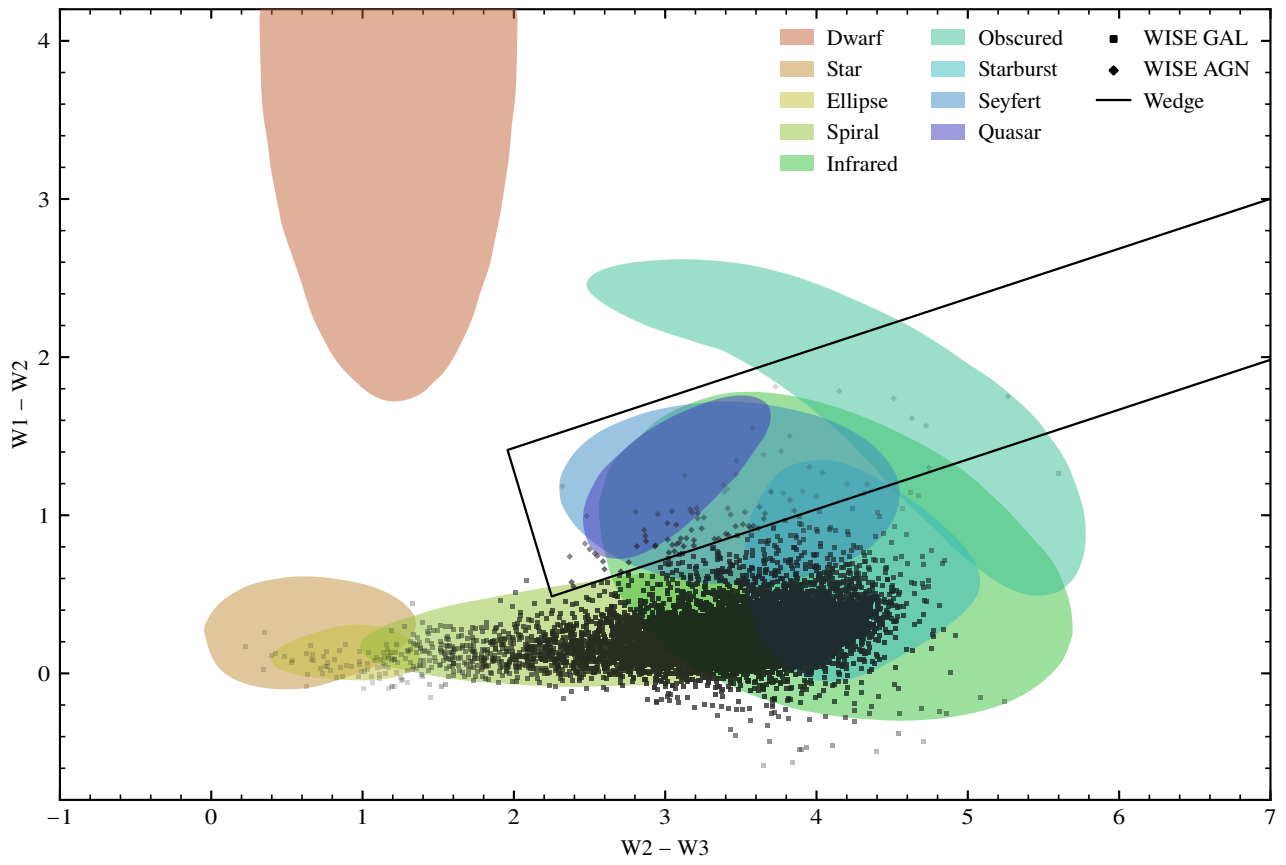


Figure 9

SDSS	1.000	0.680	0.301	0.145
BPT	0.680	1.000	0.271	0.144
WHAN	0.301	0.271	1.000	0.062
WISE	0.145	0.144	0.062	1.000
	SDSS	BPT	WHAN	WISE

Figure 10

SDSS	1.000	0.433	0.301	0.145
BPT	0.433	1.000	0.631	0.104
WHAN	0.301	0.631	1.000	0.062
WISE	0.145	0.104	0.062	1.000
	SDSS	BPT	WHAN	WISE

Figure 11

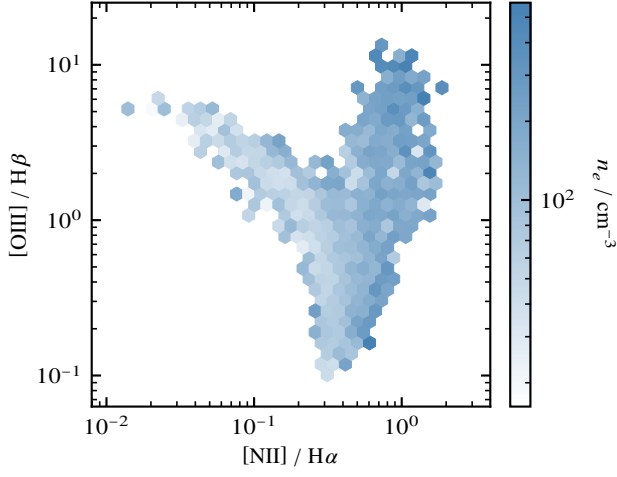


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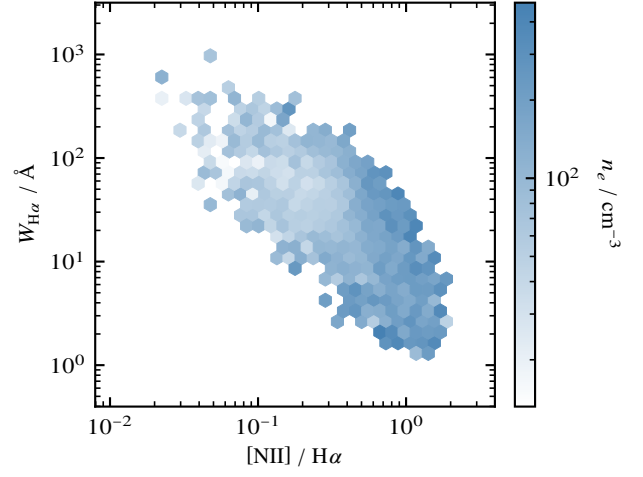


Figure 15

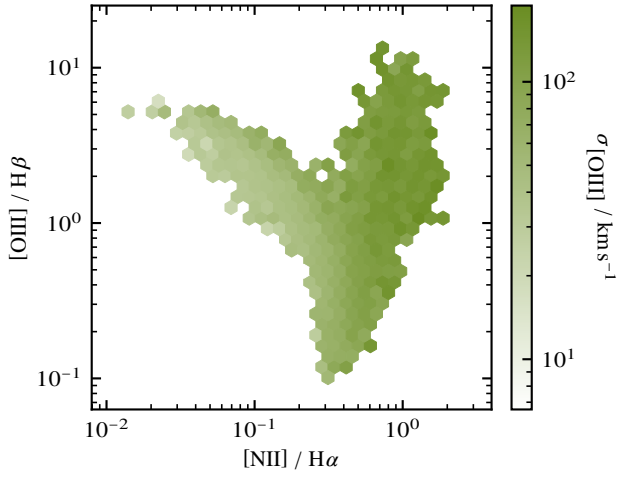


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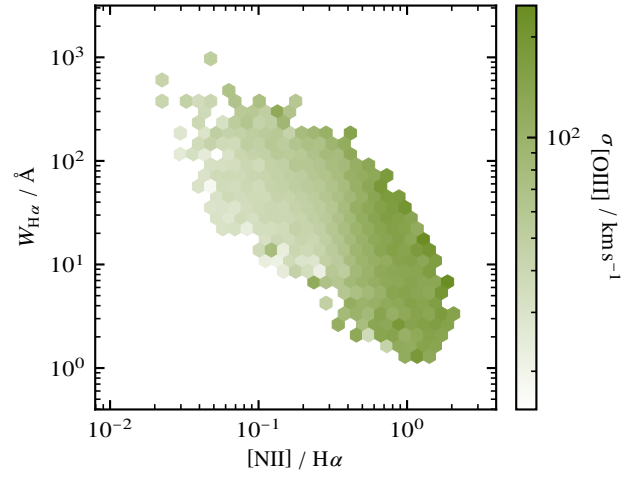


Figure 16

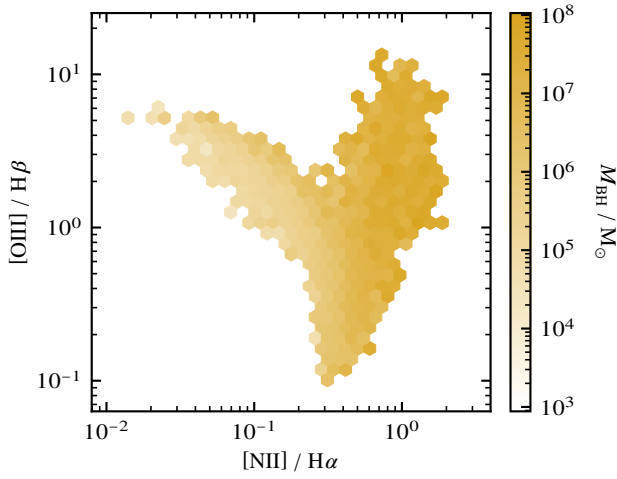


Figure 14

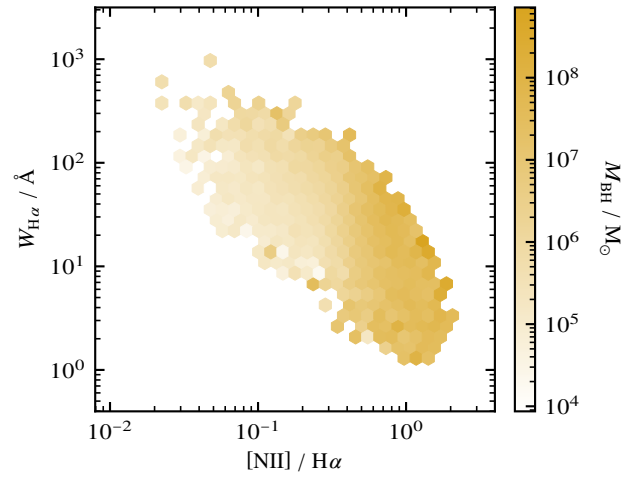


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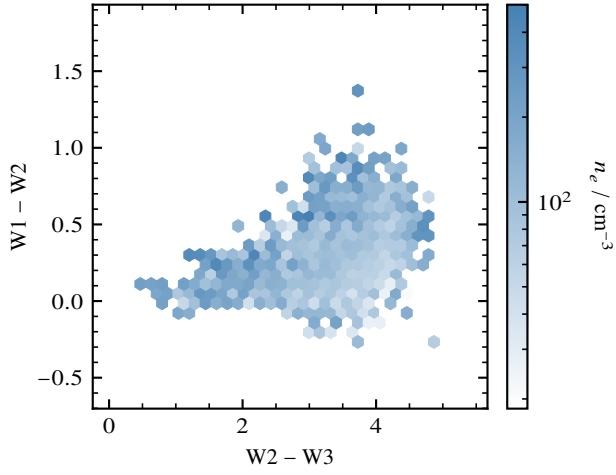


Figure 18

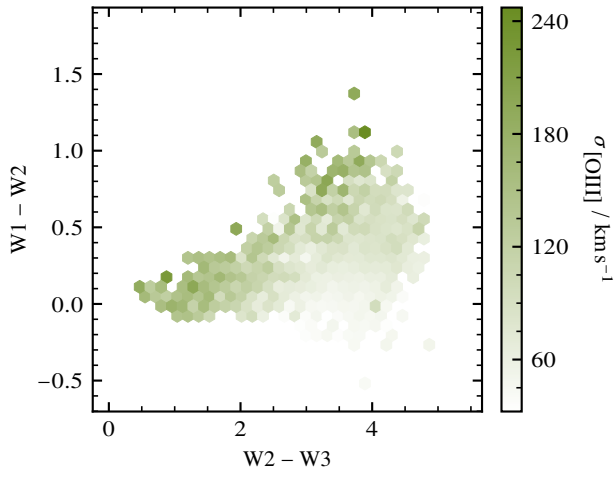


Figure 19

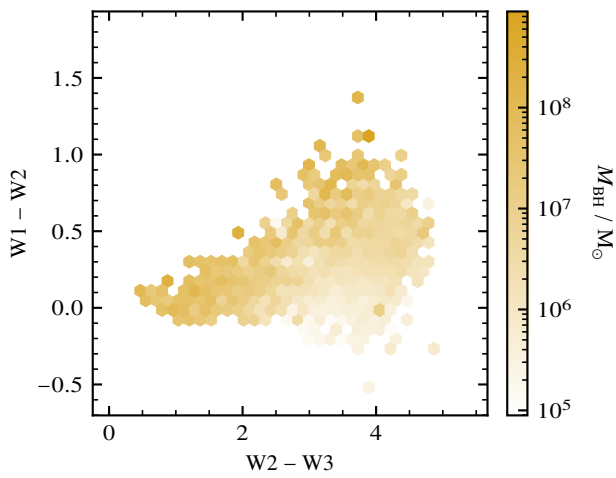


Figure 20

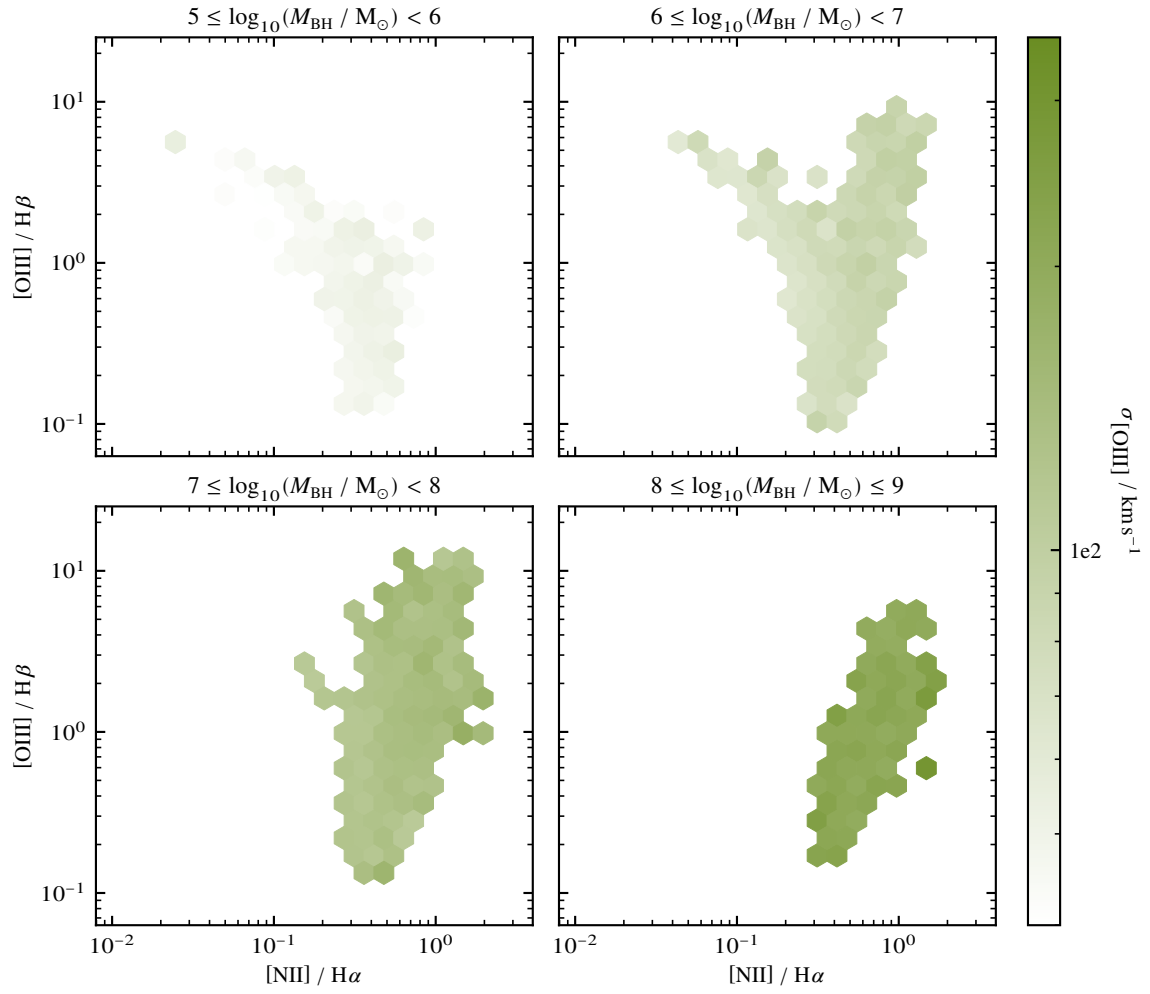


Figure 21

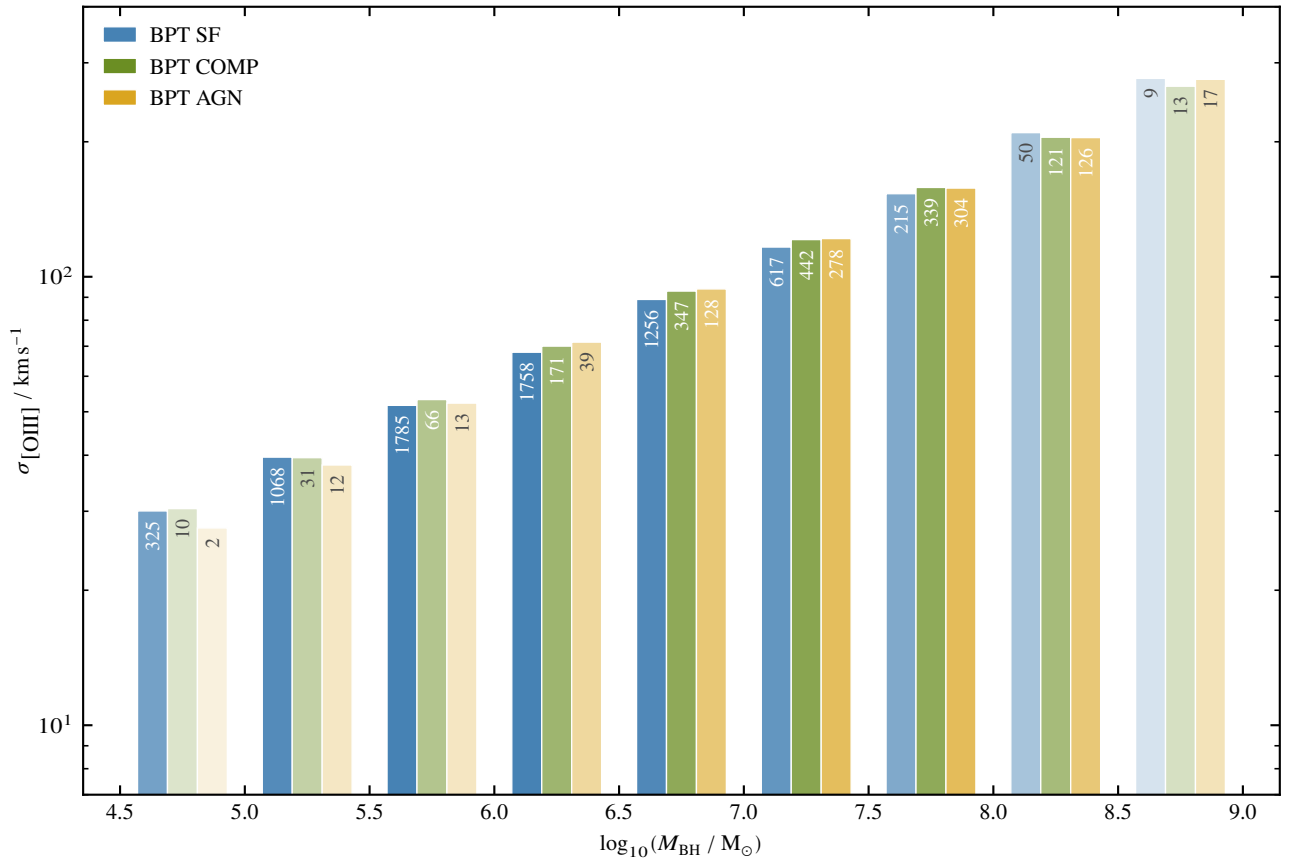


Figure 22

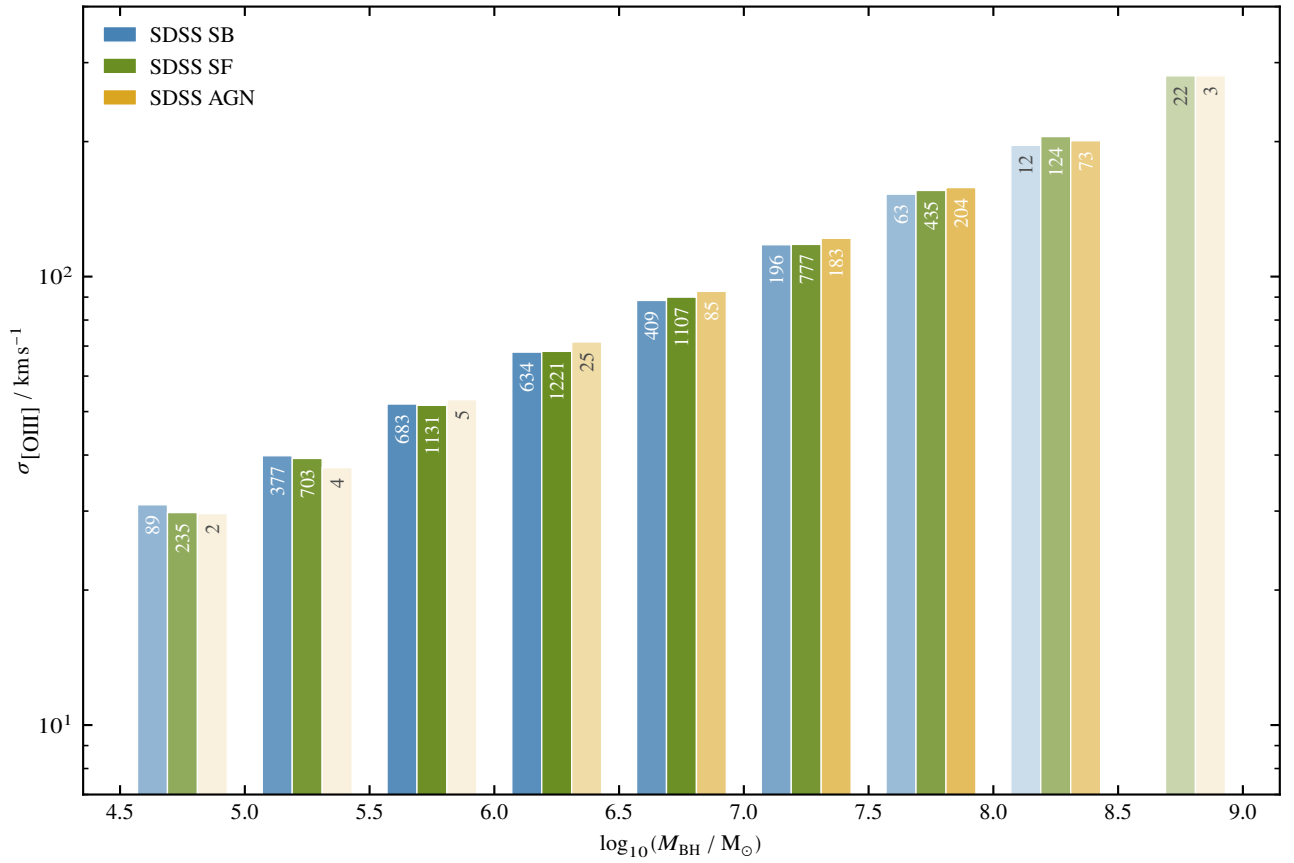


Figure 23

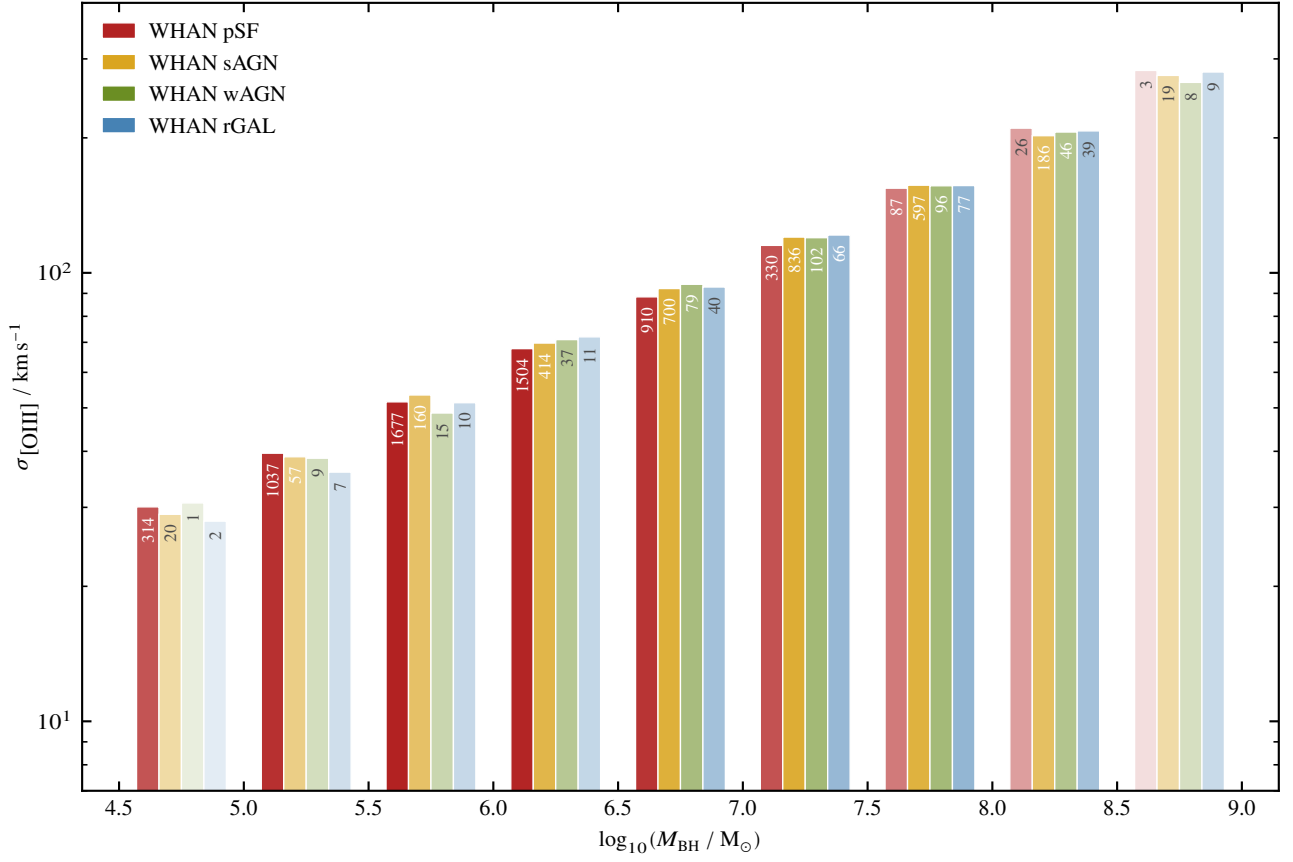


Figure 24

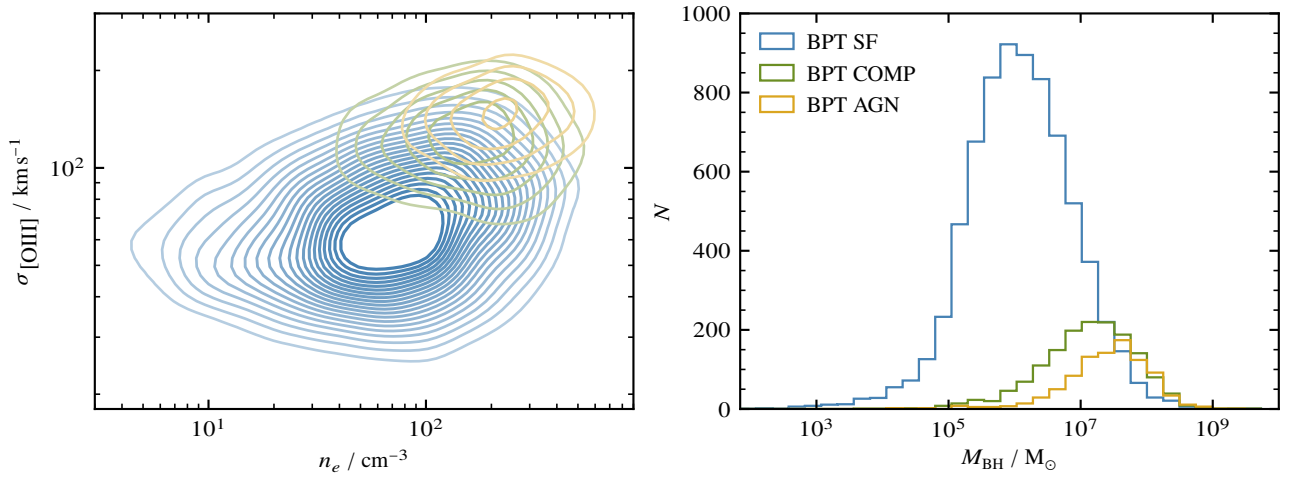


Figure 25

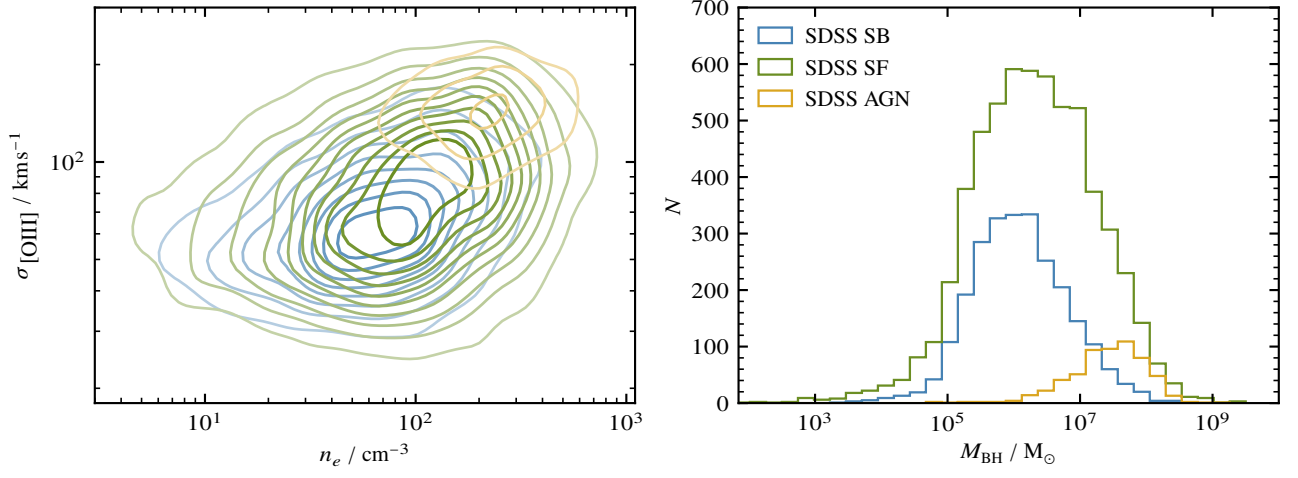


Figure 26

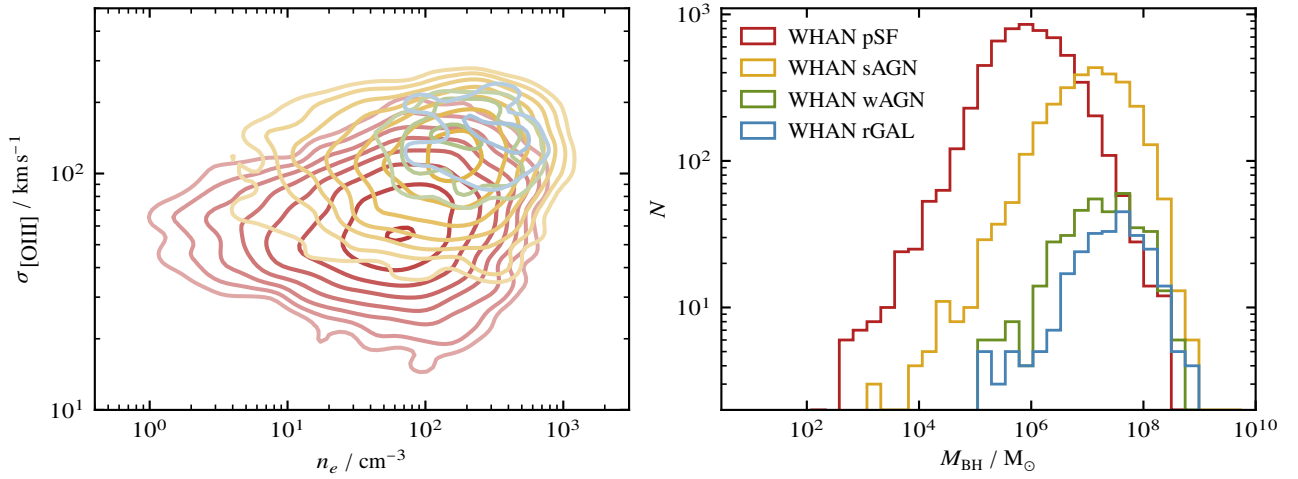


Figure 27

Appendix

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1 SELECT TOP 10000
2     g.sii_6717_flux, g.sii_6731_flux, g.nii_6584_flux, g.oi_6300_flux, g.oiii_5007_flux,
3     g.h_alpha_flux, g.h_beta_flux,
4     g.h_alpha_eqw, g.nii_6584_eqw, g.oiii_sigma,
5     w.w1mpro, w.w2mpro, w.w3mpro,
6     s.z, s.class, s.subclass, s.bestobjid
7 FROM SpecObj AS s
8 JOIN galSpecLine AS g ON s.specobjid = g.specobjid
9 JOIN PhotoTag AS p ON s.bestobjid = p.objid
10 OUTER APPLY ( SELECT TOP 1 x.wise_cntr
11                FROM wise_xmatch AS x
12                WHERE p.objid = x.sdss_objid AND x.match_dist < 3
13                ORDER BY x.match_dist ASC
14            ) AS y
15 LEFT JOIN wise_allsky AS w ON y.wise_cntr = w.cntr
16 WHERE s.class = 'GALAXY'
17 AND s.subclass != 'BROADLINE'
18 AND s.subclass != 'AGN BROADLINE'
19 AND s.subclass != 'STARFORMING BROADLINE'
20 AND s.subclass != 'STARBURST BROADLINE'
21 AND s.snmedian_r > 5.0
22 AND s.z < 0.35
23 AND g.sii_6717_eqw < 0.0
24 AND g.sii_6731_eqw < 0.0
25 AND g.nii_6584_eqw < 0.0
26 AND g.oi_6300_eqw < 0.0
27 AND g.oiii_5007_eqw < 0.0
28 AND g.h_alpha_eqw < 0.0
29 AND g.h_beta_eqw < 0.0
30 AND (2.355 * g.sigma_forbidden) < 500
31 AND (2.355 * g.sigma_balmer) < 500
32 AND g.sii_6717_flux > 3 * g.sii_6717_flux_err
33 AND g.sii_6731_flux > 3 * g.sii_6731_flux_err
34 AND g.nii_6584_flux > 3 * g.nii_6584_flux_err
35 AND g.oi_6300_flux > 3 * g.oi_6300_flux_err
36 AND g.oiii_5007_flux > 3 * g.oiii_5007_flux_err
37 AND g.h_alpha_flux > 3 * g.h_alpha_flux_err
38 AND g.h_beta_flux > 3 * g.h_beta_flux_err
39 ORDER BY s.specobjid ASC

```